WEEK 5 REPORT

Healthcare Data Modeling and Billing Analysis using Power BI

-done by **Sai Meera R** of **B.sc Data Science & AI**

Introduction

Healthcare organizations generate a large amount of data related to patients, hospital admissions, treatments, billing, and insurance. Organizing this data properly helps in understanding healthcare operations and finding useful patterns.

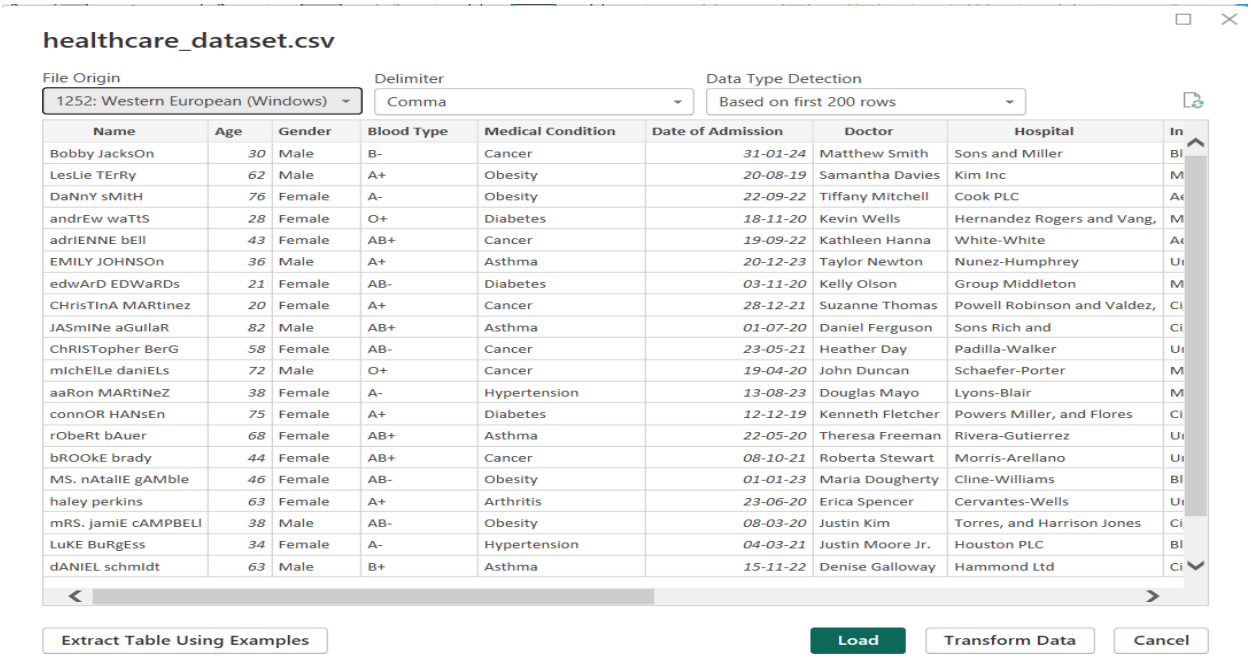
In Week 5, the healthcare dataset was transformed and structured using Power Query and Power BI. The data was divided into Dim_Patient, Dim_Admission, and Fact_Billing tables. Merge Queries were used to combine the required information, and relationships were created between the dimension and fact tables.

A separate Measure Table was also created to store important DAX measures such as Total Patients, Total Billing Amount, Average Billing Amount, Average Length of Stay, and Readmission Rate. This structured model provides a foundation for creating healthcare visualizations and dashboards.

Importing the Dataset

The healthcare dataset was imported into **Power BI** using the **Get Data** option. The dataset was loaded and opened in **Power Query Editor** for further transformation and preparation.

After importing, the available columns and data types were checked to understand the structure of the dataset before creating the required tables.



Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	In
Bobby JacksOn	30	Male	B-	Cancer	31-01-24	Matthew Smith	Sons and Miller	Bl
LesLie TErRy	62	Male	A+	Obesity	20-08-19	Samantha Davies	Kim Inc	M
DaNnY sMitH	76	Female	A-	Obesity	22-09-22	Tiffany Mitchell	Cook PLC	Ae
andrEw waTtS	28	Female	O+	Diabetes	18-11-20	Kevin Wells	Hernandez Rogers and Vang,	M
adrIENNE bEll	43	Female	AB+	Cancer	19-09-22	Kathleen Hanna	White-White	Ae
EMILY JOHNSOn	36	Male	A+	Asthma	20-12-23	Taylor Newton	Nunez-Humphrey	Ur
edwArD EDWaRDs	21	Female	AB-	Diabetes	03-11-20	Kelly Olson	Group Middleton	M
CHrisTInA MARtInez	20	Female	A+	Cancer	28-12-21	Suzanne Thomas	Powell Robinson and Valdez,	Ci
JASmINe aGullaR	82	Male	AB+	Asthma	01-07-20	Daniel Ferguson	Sons Rich and	Ci
ChRISTopher BerG	58	Female	AB-	Cancer	23-05-21	Heather Day	Padilla-Walker	Ur
mIchElLe daniELs	72	Male	O+	Cancer	19-04-20	John Duncan	Schaefer-Porter	M
aaRon MARtiNeZ	38	Female	A-	Hypertension	13-08-23	Douglas Mayo	Lyons-Blair	M
connOR HANsEn	75	Female	A+	Diabetes	12-12-19	Kenneth Fletcher	Powers Miller, and Flores	Ci
rObeRt bAuer	68	Female	AB+	Asthma	22-05-20	Theresa Freeman	Rivera-Gutierrez	Ur
bROOkE brady	44	Female	AB+	Cancer	08-10-21	Roberta Stewart	Morris-Arellano	Ur
MS. nAtalIE gAMble	46	Female	AB-	Obesity	01-01-23	Maria Dougherty	Cline-Williams	Bl
haley perkins	63	Female	A+	Arthritis	23-06-20	Erica Spencer	Cervantes-Wells	Ur
mRS. jamiE cAMPBELl	38	Male	AB-	Obesity	08-03-20	Justin Kim	Torres, and Harrison Jones	Ci
LuKE BuRgEss	34	Female	A-	Hypertension	04-03-21	Justin Moore Jr.	Houston PLC	Bl
dANIEL schmIdt	63	Male	B+	Asthma	15-11-22	Denise Galloway	Hammond Ltd	Ci

Data Preparation

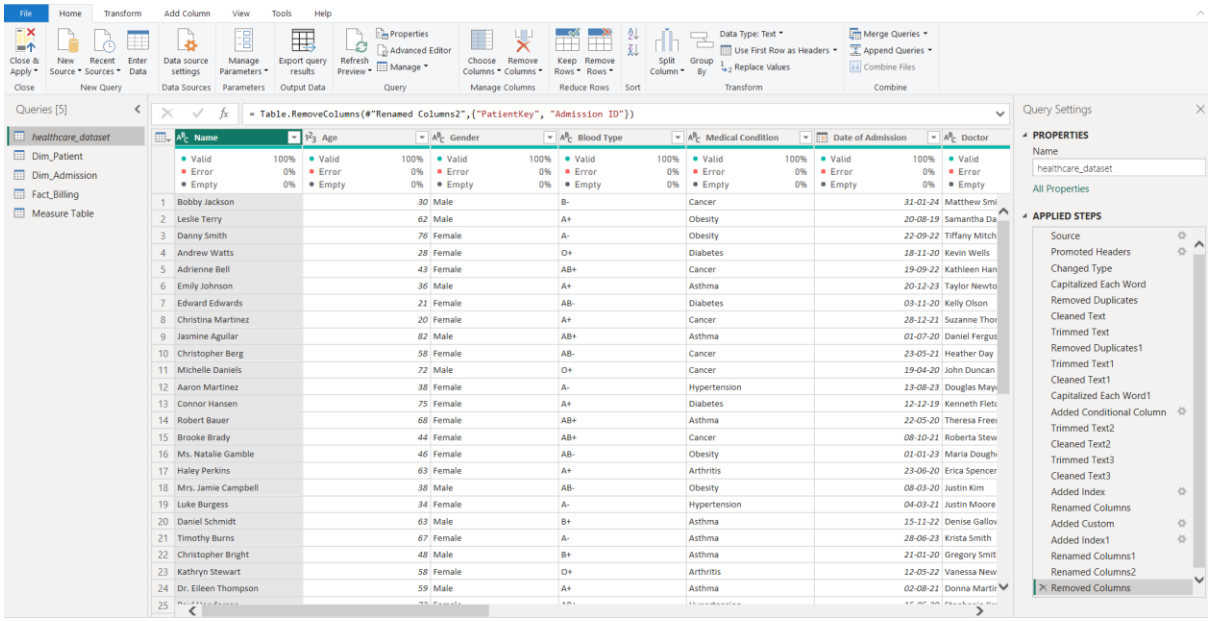
The healthcare dataset was first loaded into Power BI Power Query. The data was examined and prepared for further analysis.

The required columns were selected and organized based on their purpose. The data was divided into dimension and fact tables to create a structured model.

The main tables created were:

- **Dim_Patient** – contains patient-related information.
- **Dim_Admission** – contains admission-related information.
- **Fact_Billing** – contains billing and admission transaction details.
- **Measure Table** – stores the DAX measures used for analysis.

Tables created during data preparation:



Creating Dim_Patient

The patient information was separated into a dimension table named **Dim_Patient**.

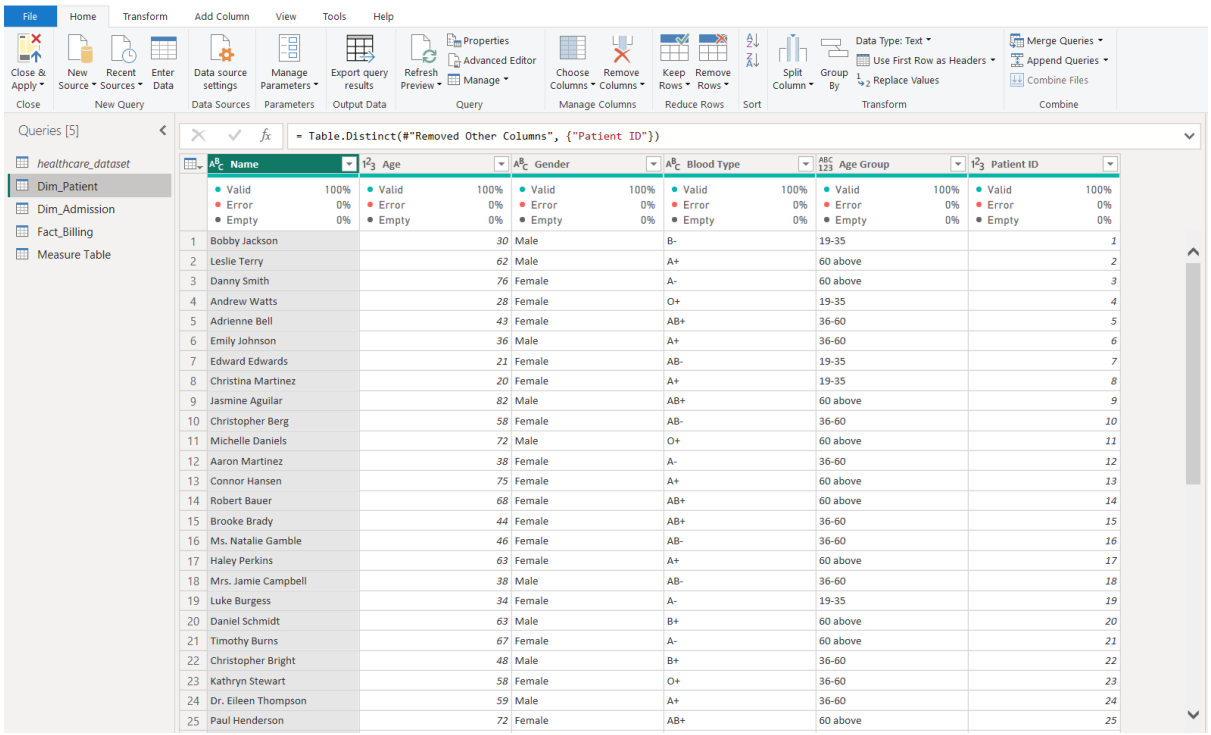
The table contains important patient attributes such as:

- Patient ID
- Name
- Age
- Age Group
- Gender

- Blood Type

Duplicate patient records were removed so that each patient is represented uniquely in the dimension table.

Dim_Patient table after data preparation:



Creating Dim_Admission

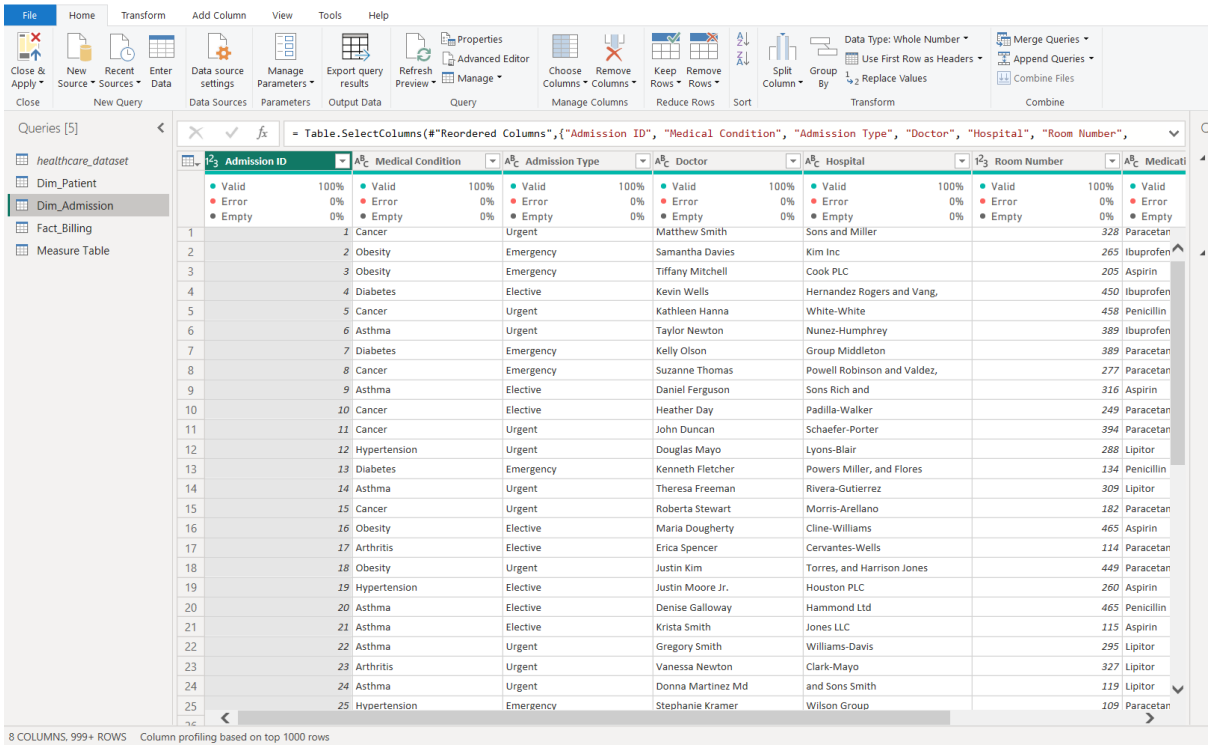
A separate **Dim_Admission** table was created to organize admission-related information.

It contains fields such as:

- Admission ID
- Admission Type
- Doctor
- Hospital
- Medical Condition
- Medication
- Room Number
- Test Results

This separation makes it easier to analyze healthcare admissions using different attributes.

Dim_Admission table:



	Admission ID	Medical Condition	Admission Type	Doctor	Hospital	Room Number	Medicati
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
1	1	Cancer	Urgent	Matthew Smith	Sons and Miller	328	Paracetar
2	2	Obesity	Emergency	Samantha Davies	Kim Inc	265	Ibuprofen
3	3	Obesity	Emergency	Tiffany Mitchell	Cook PLC	205	Aspirin
4	4	Diabetes	Elective	Kevin Wells	Hernandez Rogers and Vang,	450	Ibuprofen
5	5	Cancer	Urgent	Kathleen Hanna	White-White	458	Penicillin
6	6	Asthma	Urgent	Taylor Newton	Nunez-Humphrey	389	Ibuprofen
7	7	Diabetes	Emergency	Kelly Olson	Group Middleton	389	Paracetar
8	8	Cancer	Emergency	Suzanne Thomas	Powell Robinson and Valdez,	277	Paracetar
9	9	Asthma	Elective	Daniel Ferguson	Sons Rich and	316	Aspirin
10	10	Cancer	Elective	Heather Day	Padilla-Walker	249	Paracetar
11	11	Cancer	Urgent	John Duncan	Schaefer-Porter	394	Paracetar
12	12	Hypertension	Urgent	Douglas Mayo	Lyons-Blair	288	Lipitor
13	13	Diabetes	Emergency	Kenneth Fletcher	Powers Miller, and Flores	134	Penicillin
14	14	Asthma	Urgent	Theresa Freeman	Rivera-Gutierrez	309	Lipitor
15	15	Cancer	Urgent	Roberta Stewart	Morris-Arellano	182	Paracetar
16	16	Obesity	Elective	Maria Dougherty	Cline-Williams	465	Aspirin
17	17	Arthritis	Elective	Erica Spencer	Cervantes-Wells	114	Paracetar
18	18	Obesity	Urgent	Justin Kim	Torres, and Harrison Jones	449	Paracetar
19	19	Hypertension	Elective	Justin Moore Jr.	Houston PLC	260	Aspirin
20	20	Asthma	Elective	Denise Galloway	Hammond Ltd	465	Penicillin
21	21	Asthma	Elective	Krista Smith	Jones LLC	115	Aspirin
22	22	Asthma	Urgent	Gregory Smith	Williams-Davis	295	Lipitor
23	23	Arthritis	Urgent	Vanessa Newton	Clark-Mayo	327	Lipitor
24	24	Asthma	Urgent	Donna Martinez Md	and Sons Smith	119	Lipitor
25	25	Hypertension	Emergency	Stephanie Kramer	Wilson Group	109	Paracetar

Creating Fact_Billing

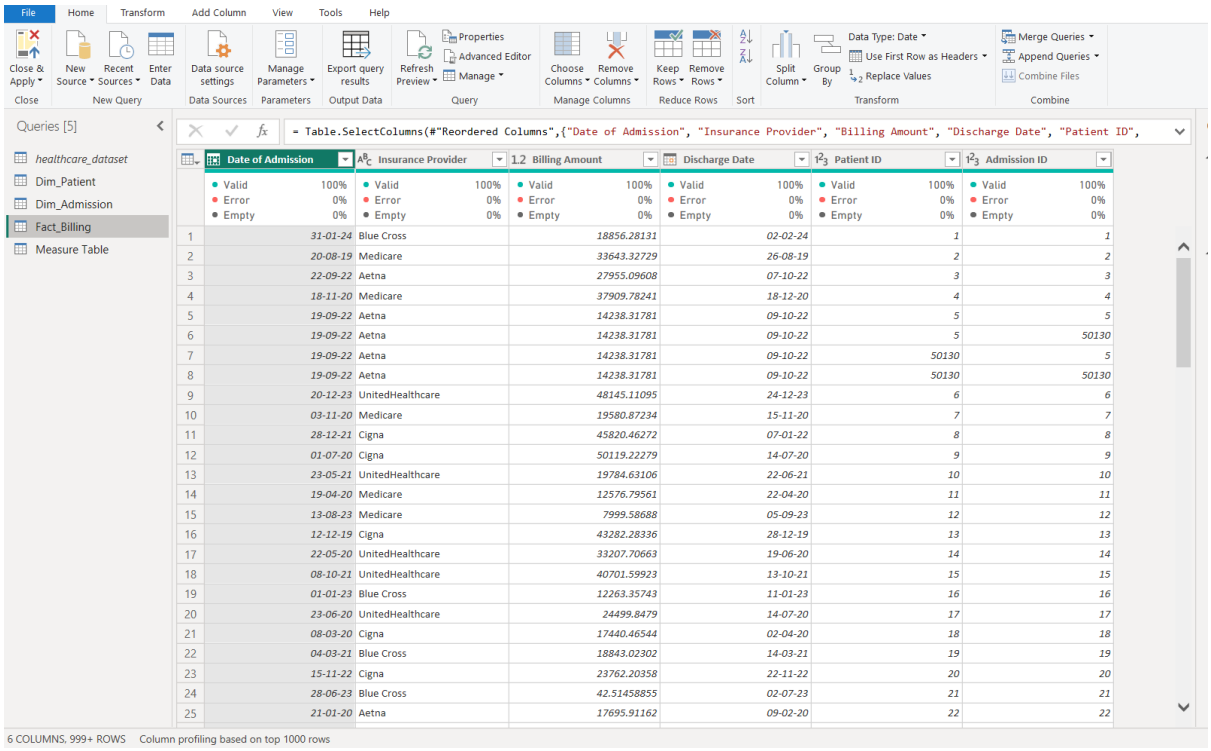
The billing-related information was organized into a fact table named **Fact_Billing**.

The table contains important fields such as:

- Admission ID
- Billing Amount
- Date of Admission
- Discharge Date
- Insurance Provider
- Patient ID

The fact table is used to store the measurable billing information and connect it with the patient and admission dimensions.

Fact_Billing table after transformation:



	Date of Admission	Insurance Provider	Billing Amount	Discharge Date	Patient ID	Admission ID
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	31-01-24	Blue Cross	18856.28131	02-02-24	1	1
2	20-08-19	Medicare	33643.32729	26-08-19	2	2
3	22-09-22	Aetna	27955.09608	07-10-22	3	3
4	18-11-20	Medicare	37909.78241	18-12-20	4	4
5	19-09-22	Aetna	14238.31781	09-10-22	5	5
6	19-09-22	Aetna	14238.31781	09-10-22	5	50130
7	19-09-22	Aetna	14238.31781	09-10-22	50130	5
8	19-09-22	Aetna	14238.31781	09-10-22	50130	50130
9	20-12-23	UnitedHealthcare	48145.11095	24-12-23	6	6
10	03-11-20	Medicare	19580.87234	15-11-20	7	7
11	28-12-21	Cigna	45820.46272	07-01-22	8	8
12	01-07-20	Cigna	50119.22279	14-07-20	9	9
13	23-05-21	UnitedHealthcare	19784.63106	22-06-21	10	10
14	19-04-20	Medicare	12576.79561	22-04-20	11	11
15	13-08-23	Medicare	7999.58688	05-09-23	12	12
16	12-12-19	Cigna	43282.28336	28-12-19	13	13
17	22-05-20	UnitedHealthcare	33207.70663	19-06-20	14	14
18	08-10-21	UnitedHealthcare	40701.59923	13-10-21	15	15
19	01-01-23	Blue Cross	12263.35743	11-01-23	16	16
20	23-06-20	UnitedHealthcare	24499.8479	14-07-20	17	17
21	08-03-20	Cigna	17440.46544	02-04-20	18	18
22	04-03-21	Blue Cross	18843.02302	14-03-21	19	19
23	15-11-22	Cigna	23762.20358	22-11-22	20	20
24	28-06-23	Blue Cross	42.51458855	02-07-23	21	21
25	21-01-20	Aetna	17695.91162	09-02-20	22	22

Merging Queries

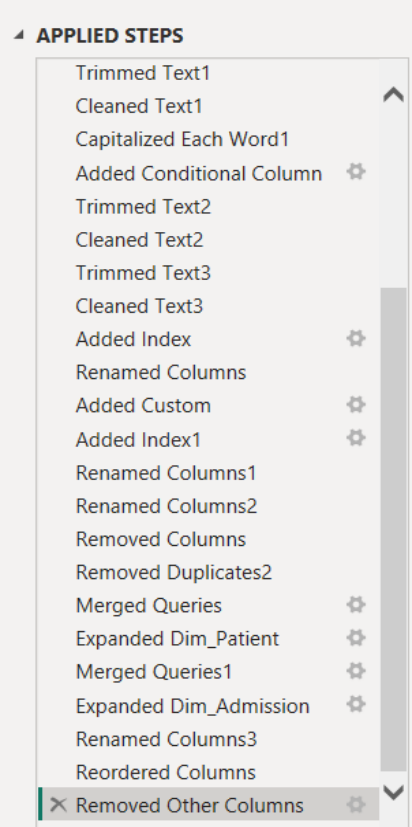
The required patient and admission information was combined with the billing data using **Merge Queries** in Power Query.

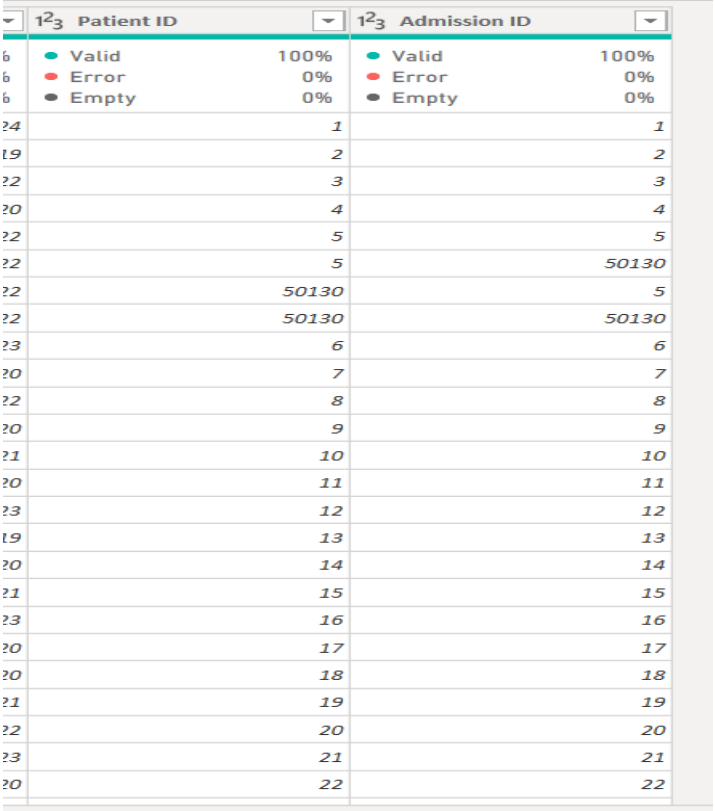
The tables were merged using their corresponding key fields, such as **Patient ID** and **Admission ID**.

After merging, the required columns were expanded and included in the final Fact_Billing table.

This helped create a more complete fact table containing both billing information and related healthcare details.

Merging and expanding dimension tables in Fact_Billing:





	Patient ID		Admission ID	
	Valid	100%	Valid	100%
	Error	0%	Error	0%
	Empty	0%	Empty	0%
24	1		1	
19	2		2	
22	3		3	
20	4		4	
22	5		5	
22	5		50130	
22	50130		5	
22	50130		50130	
23	6		6	
20	7		7	
22	8		8	
20	9		9	
21	10		10	
20	11		11	
23	12		12	
19	13		13	
20	14		14	
21	15		15	
23	16		16	
20	17		17	
20	18		18	
21	19		19	
22	20		20	
23	21		21	
20	22		22	

Data Model and Relationships

The final data model follows a **star-schema structure**.

The relationships were created between the dimension tables and the fact table:

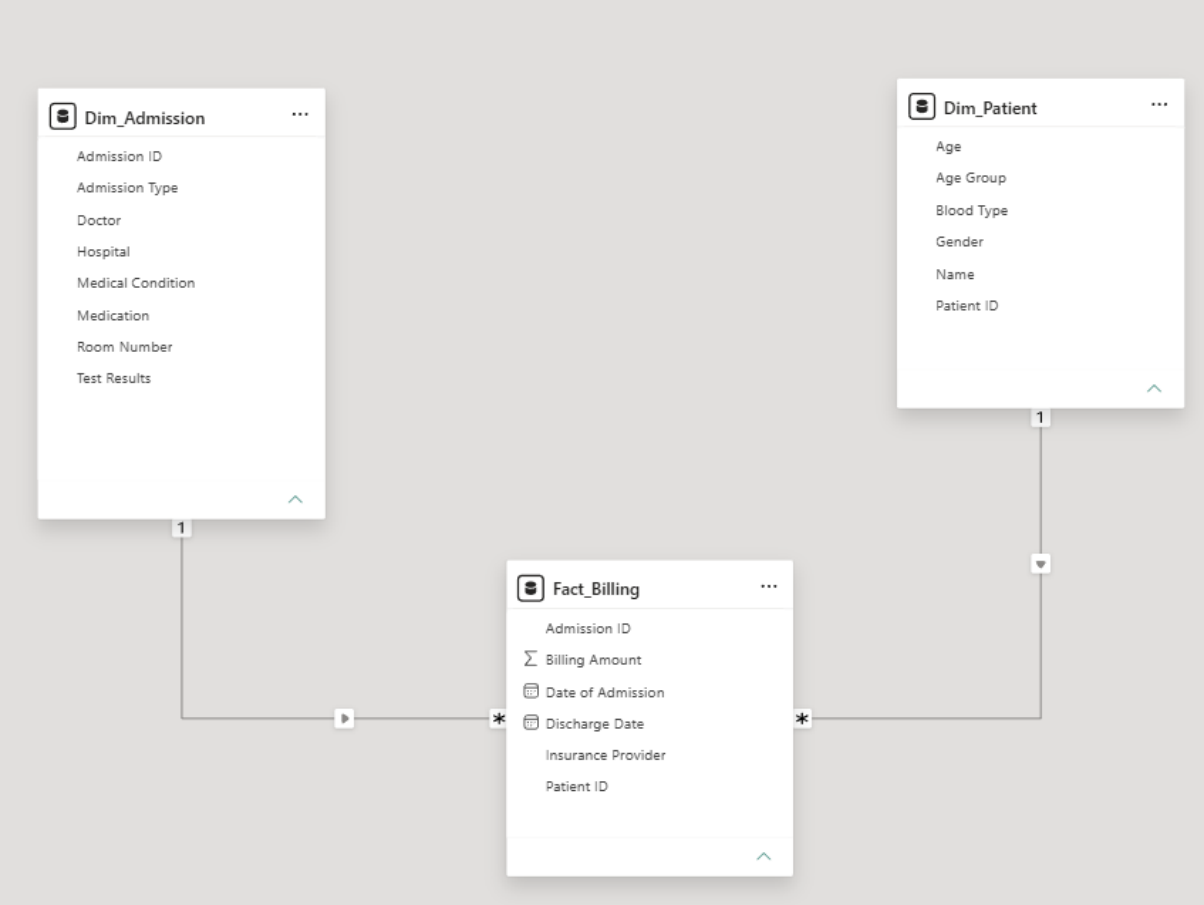
- Dim_Admission[Admission ID] → Fact_Billing[Admission ID]
- Dim_Patient[Patient ID] → Fact_Billing[Patient ID]

The dimension tables are on the **one side (1)**, while Fact_Billing is on the **many side (*)**.

The relationships use **single-direction filtering**.

The original healthcare_dataset query was kept as a source for transformations but was disabled from loading into the final model.

Relationships between dimension and fact tables:



Creating the Measure Table

A separate **Measure Table** was created to organize the DAX calculations.

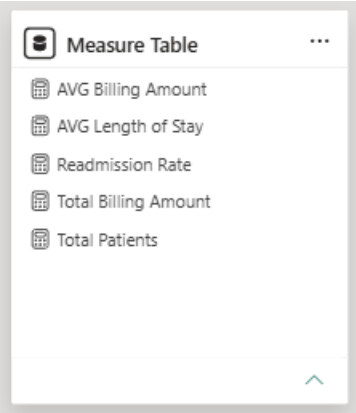
The following measures were created:

1. Total Patients

2. Total Billing Amount
3. Average Billing Amount
4. Average Length of Stay
5. Readmission Rate

Keeping the measures in a separate table makes the model easier to understand and keeps the calculations organized.

Measure Table containing DAX calculations:



DAX Measures

- **Total Patients**

Total Patients = COUNTROWS(Fact_Billing)

This measure calculates the total number of records/patient billing entries in the fact table.

- **Total Billing Amount**

Total Billing Amount =

SUM(Fact_Billing[Billing Amount])

This calculates the total billing amount across the dataset.

- **Average Billing Amount**

AVG Billing Amount =

AVERAGE(Fact_Billing[Billing Amount])

This calculates the average billing amount per record.

- **Average Length of Stay**

AVG Length of Stay =

AVERAGEX(

```
    Fact_Billing,
    DATEDIFF(
        Fact_Billing[Date of Admission],
        Fact_Billing[Discharge Date],
        DAY
    )
)
```

This calculates the average number of days patients stayed in the hospital.

- **Readmission Rate**

```
Readmission Rate =
VAR TotalCount = [Total Patients]

VAR RepeatPatients =
    COUNTROWS(
        FILTER(
            SUMMARIZE(
                Fact_Billing,
                Fact_Billing[Patient ID],
                "AdmissionCount",
                COUNTROWS(Fact_Billing)
            ),
            [AdmissionCount] > 1
        )
    )

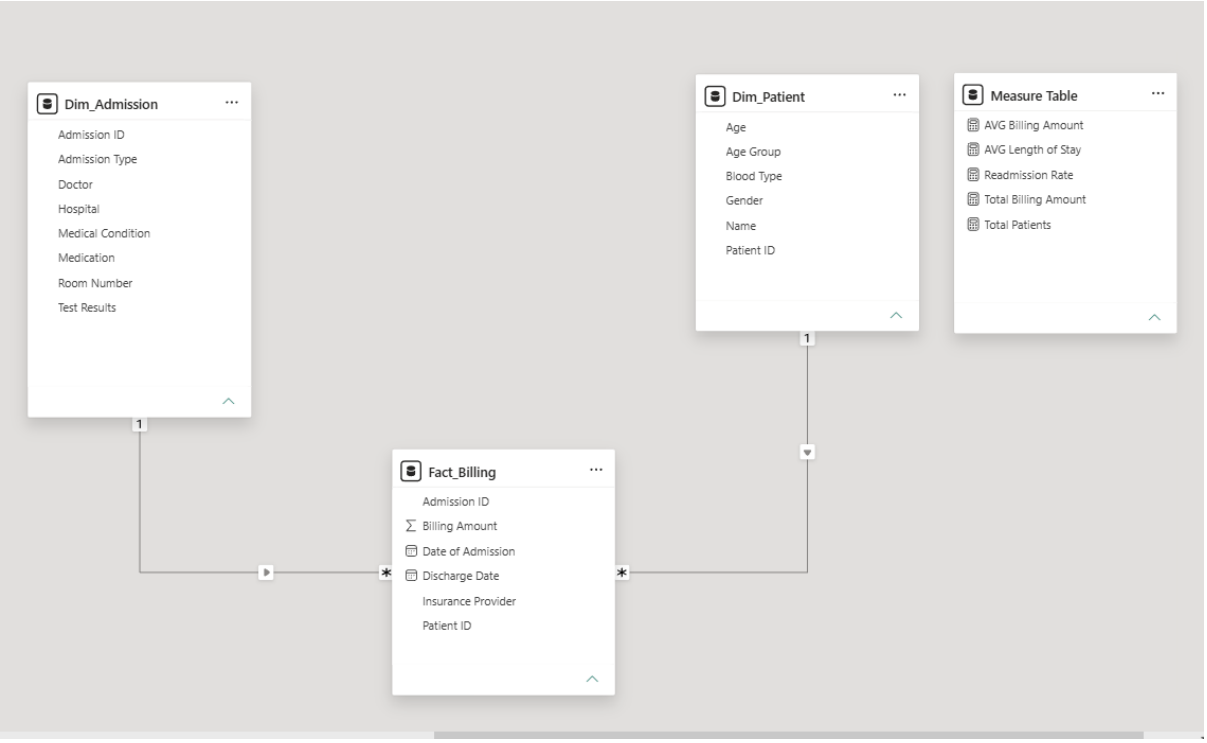
RETURN
DIVIDE(RepeatPatients, TotalCount, 0)
```

This measure calculates the proportion of patients who have more than one admission record.

Final Model

The completed Week 5 model contains the required dimension tables, fact table, relationships, and measure table.

The model allows healthcare information to be analyzed based on patient details, admission information, billing amounts, length of stay, and readmission patterns.



Conclusion

Week 5 focused on transforming the healthcare dataset into a structured Power BI data model. Dimension tables and a fact table were created, relevant queries were merged, relationships were established, and DAX measures were developed.

The final model provides a foundation for creating healthcare dashboards and visualizations in the next stage of the project.